Kubernetes Task - 01



Start Minikube----- minikube start --driver=docker --cpus=2 --memory=2048

```
PS C:\WINDOWS\system32> minikube start --driver=docker --cpus=2 --memory=2048
* minikube v1.39.0 on Microsoft Windows 11 Home Single Language 26H2
* Using the docker driver based on user configuration
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.51 ...
* Downloading Kubernetes v1.37.0 preload ...
    > preloaded-images-k8s-v18-v1...:  88.12 MiB / 347.58 MiB  25.35% 4.13 MiB
```

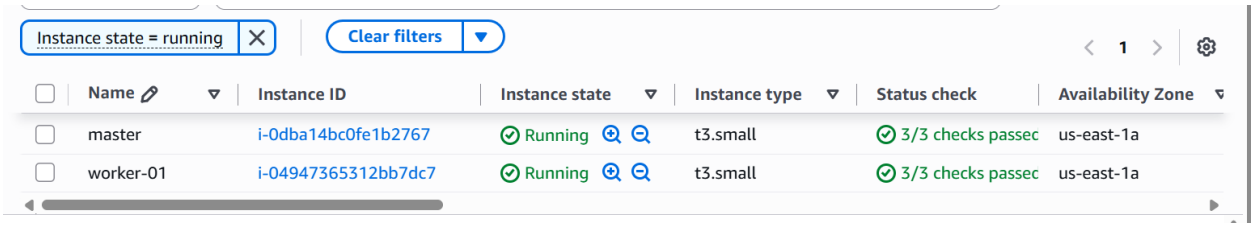
```
PS C:\WINDOWS\system32> minikube start --driver=docker --cpus=2 --memory=2048
* minikube v1.39.0 on Microsoft Windows 11 Home Single Language 26H2
* Using the docker driver based on user configuration
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.51 ...
* Downloading Kubernetes v1.37.0 preload ...
    > gcr.io/k8s-minikube/kicbase:  507.61 MiB / 507.61 MiB  100.00% 5.79 MiB p
    > preloaded-images-k8s-v18-v1...:  347.58 MiB / 347.58 MiB  100.00% 2.86 Mi
* Creating docker container (CPUs=2, Memory=2048MB) ...
* Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
* Configuring CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\WINDOWS\system32>
```

Minikube setup is complete.

Your local Kubernetes cluster is running, and `kubectl` is configured to communicate with it.

```
PS C:\WINDOWS\system32> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

<u>Kubernetes Master + 2 Worker Nodes on Ubuntu</u>



Install containerd

```
ubuntu@master:~$ sudo apt install -y containerd
Installing:
  containerd

Installing dependencies:
  runc

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 56
  Download size: 37.9 MB
  Space needed: 142 MB / 17.2 GB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 runc amd64 1.4.0-0ubuntu1 [9829 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 containerd amd64 2.2.2-0ubuntu1.2 [28.1 MB]
Fetched 37.9 MB in 0s (94.3 MB/s)
Selecting previously unselected package runc.
(Reading database ... 85362 files and directories currently installed.)
Preparing to unpack .../runc_1.4.0-0ubuntu1_amd64v3.deb ...
Unpacking runc (1.4.0-0ubuntu1) ...
Selecting previously unselected package containerd.
Preparing to unpack .../containerd_2.2.2-0ubuntu1.2_amd64v3.deb ...
Unpacking containerd (2.2.2-0ubuntu1.2) ...
Setting up runc (1.4.0-0ubuntu1) ...
Setting up containerd (2.2.2-0ubuntu1.2) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/containerd.service' → '/usr/lib/systemd/system/containerd.service'.
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
```

```
ubuntu@master:~$ containerd --version
containerd github.com/containerd/containerd/v2 2.2.2
```

```
ubuntu@master:~$ sudo systemctl status containerd
● containerd.service - containerd container runtime
     Loaded: loaded (/usr/lib/systemd/system/containerd.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-10 12:03:40 UTC; 1min 20s ago
 Invocation: 90e630eab78e45cfa4e762b4d97266e6
       Docs: https://containerd.io
    Process: 1922 ExecStartPre=/sbin/modprobe overlay (code=exited, status=0/SUCCESS)
   Main PID: 1924 (containerd)
      Tasks: 8
     Memory: 14M (peak: 16.9M)
        CPU: 191ms
     CGroup: /system.slice/containerd.service
             └─1924 /usr/bin/containerd
```

Disable swap

```
ubuntu@master:~$ sudo swapoff -a
ubuntu@master:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       420Mi       887Mi       2.8Mi       768Mi       1.4Gi
Swap:             0B          0B          0B
ubuntu@master:~$
```

Enable Kubernetes kernel modules

```
ubuntu@master:~$ sudo modprobe overlay
sudo modprobe br_netfilter
ubuntu@master:~$ lsmod | grep -E 'overlay|br_netfilter'
br_netfilter           32768  0
bridge                430080  1 br_netfilter
overlay               233472  0
ubuntu@master:~$ ^C
ubuntu@master:~$
```

Both required kernel modules are loaded: br_netfilter

bridge

Overlay

Configure Kubernetes networking

```
ubuntu@master:~$ sudo tee /etc/sysctl.d/kubernetes.conf <<EOF
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
EOF
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
ubuntu@master:~$ sudo sysctl --system
* Applying /usr/lib/sysctl.d/10-apparmor.conf ...
* Applying /usr/lib/sysctl.d/10-coredump-debian.conf ...
* Applying /etc/sysctl.d/50-cloudimg-settings.conf ...
* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
* Applying /usr/lib/sysctl.d/55-bufferbloat.conf ...
* Applying /usr/lib/sysctl.d/55-console-messages.conf ...
* Applying /usr/lib/sysctl.d/55-ipv6-privacy.conf ...
* Applying /usr/lib/sysctl.d/55-kernel-hardening.conf ...
* Applying /usr/lib/sysctl.d/55-magic-sysrq.conf ...
* Applying /usr/lib/sysctl.d/55-map-count.conf ...
* Applying /usr/lib/sysctl.d/55-network-security.conf ...
* Applying /usr/lib/sysctl.d/55-ptrace.conf ...
* Applying /usr/lib/sysctl.d/55-zeropage.conf ...
* Applying /etc/sysctl.d/99-cloudimg-ipv6.conf ...
* Applying /etc/sysctl.d/kubernetes.conf ...
kernel.apparmor_restrict_unprivileged_userns = 1
kernel.apparmor_restrict_unprivileged_unconfined = 1
kernel.core_pattern = core
net.ipv4.neigh.default.gc_thresh2 = 15360
net.ipv4.neigh.default.gc_thresh3 = 16384
net.netfilter.nf_conntrack_max = 1048576
kernel.pid_max = 4194304
net.core.default_qdisc = fq_codel
kernel.printk = 4 4 1 7
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
kernel.kptr_restrict = 1
kernel.sysrq = 176
vm.max_map_count = 1048576
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.all.rp_filter = 2
kernel.yama.ptrace_scope = 1
vm.mmap_min_addr = 65536
net.ipv6.conf.all.use_tempaddr = 0
net.ipv6.conf.default.use_tempaddr = 0
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
ubuntu@master:~$ sysctl net.ipv4.ip_forward
net.ipv4.ip_forward = 1
ubuntu@master:~$ sysctl net.bridge.bridge-nf-call-iptables
net.bridge.bridge-nf-call-iptables = 1
ubuntu@master:~$ sysctl net.ipv4.ip_forward
net.ipv4.ip_forward = 1
ubuntu@master:~$ ^C
ubuntu@master:~$
```

Install Kubernetes packages

We'll install:

kubeadm → creates/initializes the Kubernetes cluster

kubelet → runs Kubernetes components on the node

kubectl → command-line tool to manage the cluster

```
ubuntu@master:~$ sudo apt-get update
sudo apt-get install -y apt-transport-https ca-certificates curl gpg
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.18.0-1ubuntu2.8).
curl set to manually installed.
gpg is already the newest version (2.4.8-4ubuntu3.1).
gpg set to manually installed.
Solving dependencies... Done
The following NEW packages will be installed:
  apt-transport-https
The following packages will be upgraded:
  apt ca-certificates libapt-pkg7.0
3 upgraded, 1 newly installed, 0 to remove and 53 not upgraded.
```

Then add the Kubernetes signing key:

```
ubuntu@master:~$ sudo mkdir -p -m 755 /etc/apt/keyrings
ubuntu@master:~$ curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.34/deb/Release.key | \
sudo gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
ubuntu@master:~$
```

Then add the Kubernetes repository:

```
ubuntu@master:~$ echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.34/deb/ /' | \
sudo tee /etc/apt/sources.list.d/kubernetes.list
deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.34/deb/ /
```

Then install:

```
ubuntu@master:~$ sudo apt-get install -y kubelet kubeadm kubectl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following additional packages will be installed:
  kubernetes-cni
The following NEW packages will be installed:
  kubeadm kubectl kubelet kubernetes-cni
0 upgraded, 4 newly installed, 0 to remove and 53 not upgraded.
Need to get 77.9 MB of archives.
After this operation, 295 MB of additional disk space will be used.
Get:1 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.34/deb  kubeadm 1.34.13-1.1 [12.7 MB]
Get:2 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.34/deb  kubectl 1.34.13-1.1 [12.0 MB]
Get:3 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.34/deb  kubernetes-cni 1.7.1-1.1 [39.9 MB]
Get:4 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.34/deb  kubelet 1.34.13-1.1 [13.3 MB]
Fetched 77.9 MB in 1s (100 MB/s)
Selecting previously unselected package kubeadm.
(Reading database ... 85430 files and directories currently installed.)
Preparing to unpack .../kubeadm_1.34.13-1.1_amd64.deb ...
Unpacking kubeadm (1.34.13-1.1) ...
Selecting previously unselected package kubectl.
```

```
ubuntu@master:~$ kubeadm version
kubeadm version: &version.Info{Major:"1", Minor:"34", EmulationMajor:"", EmulationMinor:"", MinCompatibilityMajor:"", MinCompatibilityMinor:"", GitVersion:"v1.34.13", GitCo
mmit:"5bafe0f04b114c29e7bda3bd2d08919a8457b8d7", GitTreeState:"clean", BuildDate:"2026-10-27T12:05:17Z", GoVersion:"go1.26.10", Compiler:"gc", Platform:"linux/amd64"}
ubuntu@master:~$ kubectl version --client
Client Version: v1.34.13
Kustomize Version: v5.7.1
ubuntu@master:~$ kubeadm version
kubeadm version: &version.Info{Major:"1", Minor:"34", EmulationMajor:"", EmulationMinor:"", MinCompatibilityMajor:"", MinCompatibilityMinor:"", GitVersion:"v1.34.13", GitCo
mmit:"5bafe0f04b114c29e7bda3bd2d08919a8457b8d7", GitTreeState:"clean", BuildDate:"2026-10-27T12:05:17Z", GoVersion:"go1.26.10", Compiler:"gc", Platform:"linux/amd64"}
ubuntu@master:~$
```

```
            IoGid = 0
            IoUid = 0
            NoNewKeyring = false
            Root = ''
            ShimCgroup = ''
            SystemdCgroup = true

    [plugins.'io.containerd.cri.v1.runtime'.cni]
      bin_dir = ''
      bin_dirs = ['/opt/cni/bin']
      conf_dir = '/etc/cni/net.d'
      max_conf_num = 1
      setup_serially = false
```

change systemcgroup to true

```
ubuntu@master:~$ sudo nano /etc/containerd/config.toml
ubuntu@master:~$ sudo systemctl restart containerd
ubuntu@master:~$ sudo systemctl status containerd
● containerd.service - containerd container runtime
     Loaded: loaded (/usr/lib/systemd/system/containerd.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-10 12:21:51 UTC; 6s ago
 Invocation: 5afe0fc922884d8ea69ea3164a44269c
       Docs: https://containerd.io
    Process: 7624 ExecStartPre=/sbin/modprobe overlay (code=exited, status=0/SUCCESS)
   Main PID: 7626 (containerd)
      Tasks: 7
     Memory: 14M (peak: 16.5M)
        CPU: 95ms
```

Now we can initialize the Kubernetes master/control-plane.

Initialize the cluster

```
ubuntu@master:~$ sudo kubeadm init --pod-network-cidr=192.168.0.0/16
I1110 12:23:13.232712    7696 version.go:260] remote version is much newer: v1.37.2; falling back to: stable-1.34
[init] Using Kubernetes version: v1.34.13
[preflight] Running pre-flight checks
[preflight] Pulling images required for setting up a Kubernetes cluster
[preflight] This might take a minute or two, depending on the speed of your internet connection
[preflight] You can also perform this action beforehand using 'kubeadm config images pull'
[certs] Using certificateDir folder "/etc/kubernetes/pki"
[certs] Generating "ca" certificate and key
[certs] Generating "apiserver" certificate and key
[certs] apiserver serving cert is signed for DNS names [kubernetes kubernetes.default kubernetes.default.svc kubernetes.default.svc.cluster.local master] and IP
[10.30.1.170]
[certs] Generating "apiserver-kubelet-client" certificate and key
[certs] Generating "front-proxy-ca" certificate and key
[certs] Generating "front-proxy-client" certificate and key
[certs] Generating "etcd/ca" certificate and key
[certs] Generating "etcd/server" certificate and key
[certs] etcd/server serving cert is signed for DNS names [localhost master] and IPs [10.30.1.170 127.0.0.1 ::1]
```

kubeadm `init` completed successfully, the master/control-plane is initialized.

Configure `kubectl` on the master

```
ubuntu@master:~$ mkdir -p $HOME/.kube
ubuntu@master:~$ sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
ubuntu@master:~$ sudo chown $(id -u):$(id -g) $HOME/.kube/config
ubuntu@master:~$ kubectl get nodes
NAME     STATUS     ROLES           AGE   VERSION
master   NotReady   control-plane   79s   v1.34.13
ubuntu@master:~$
```

Install Calico networking

```
ubuntu@master:~$ kubectl apply -f https://raw.githubusercontent.com/projectcalico/calico/v3.30.3/manifests/calico.yaml
poddisruptionbudget.policy/calico-kube-controllers created
serviceaccount/calico-kube-controllers created
serviceaccount/calico-node created
serviceaccount/calico-cni-plugin created
configmap/calico-config created
customresourcedefinition.apiextensions.k8s.io/bgpconfigurations.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/bgpfilters.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/bgppeers.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/blockaffinities.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/caliconodestatuses.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/clusterinformations.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/felixconfigurations.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/globalnetworkpolicies.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/globalnetworksets.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/hostendpoints.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/ipamblocks.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/ipamconfigs.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/ipamhandles.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/ippools.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/ipreservations.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/kubecontrollersconfigurations.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/networkpolicies.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/networksets.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/stagedglobalnetworkpolicies.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/stagedkubernetesnetworkpolicies.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/stagednetworkpolicies.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/tiers.crd.projectcalico.org created
customresourcedefinition.apiextensions.k8s.io/adminnetworkpolicies.policy.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/baselineadminnetworkpolicies.policy.networking.k8s.io created
```

```
ubuntu@master:~$ kubectl get nodes
NAME     STATUS   ROLES           AGE     VERSION
master   Ready    control-plane   2m19s   v1.34.13
ubuntu@master:~$ ^C
ubuntu@master:~$
```

Kubernetes master/control-plane is ready.

Go to Worker 1

Install containerd

```
ubuntu@worker-01:~$ sudo apt install -y containerd
Installing:
  containerd

Installing dependencies:
  runc

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 56
  Download size: 37.9 MB
  Space needed: 142 MB / 17.1 GB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 runc amd64 1.4.0-0ubuntu1 [9829 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 containerd amd64 2.2.2-0ubuntu1.2 [28.1 MB]
Fetched 37.9 MB in 0s (97.2 MB/s)
Selecting previously unselected package runc.
(Reading database ... 85362 files and directories currently installed.)
Preparing to unpack .../runc_1.4.0-0ubuntu1_amd64v3.deb ...
Unpacking runc (1.4.0-0ubuntu1) ...
Selecting previously unselected package containerd.
Preparing to unpack .../containerd_2.2.2-0ubuntu1.2_amd64v3.deb ...
Unpacking containerd (2.2.2-0ubuntu1.2) ...
Setting up runc (1.4.0-0ubuntu1) ...
Setting up containerd (2.2.2-0ubuntu1.2) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/containerd.service' → '/usr/lib/systemd/system/containerd.service'.
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.
```

```
            NoNewKeyring = false
            Root = ''
            ShimCgroup = ''
            SystemdCgroup = true

    [plugins.'io.containerd.cri.v1.runtime'.cni]
      bin_dir = ''
      bin_dirs = ['/opt/cni/bin']
      conf_dir = '/etc/cni/net.d'
      max_conf_num = 1
      setup_serially = false
      conf_template = ''
      ip_pref = ''
      use_internal_loopback = false

  [plugins.'io.containerd.differ.v1.erofs']
    mkfs_options = []
    enable_tar_index = false
```

```
ubuntu@worker-01:~$ sudo nano /etc/containerd/config.toml
ubuntu@worker-01:~$ sudo systemctl restart containerd
ubuntu@worker-01:~$ sudo swapoff -a
ubuntu@worker-01:~$ sudo modprobe overlay
sudo modprobe br_netfilter
ubuntu@worker-01:~$ sudo tee /etc/sysctl.d/k8s.conf <<EOF
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
EOF
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
ubuntu@worker-01:~$ sudo sysctl --system
* Applying /usr/lib/sysctl.d/10-apparmor.conf ...
* Applying /usr/lib/sysctl.d/10-coredump-debian.conf ...
* Applying /etc/sysctl.d/50-cloudimg-settings.conf ...
* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
* Applying /usr/lib/sysctl.d/55-bufferbloat.conf ...
* Applying /usr/lib/sysctl.d/55-console-messages.conf ...
* Applying /usr/lib/sysctl.d/55-ipv6-privacy.conf ...
* Applying /usr/lib/sysctl.d/55-kernel-hardening.conf ...
* Applying /usr/lib/sysctl.d/55-magic-sysrq.conf ...
* Applying /usr/lib/sysctl.d/55-map-count.conf ...
* Applying /usr/lib/sysctl.d/55-network-security.conf ...
* Applying /usr/lib/sysctl.d/55-ptrace.conf ...
* Applying /usr/lib/sysctl.d/55-zeropage.conf ...
* Applying /etc/sysctl.d/99-cloudimg-ipv6.conf ...
* Applying /etc/sysctl.d/k8s.conf ...
kernel.apparmor_restrict_unprivileged_userns = 1
kernel.apparmor_restrict_unprivileged_unconfined = 1
kernel.core_pattern = core
net.ipv4.neigh.default.gc_thresh2 = 15360
net.ipv4.neigh.default.gc_thresh3 = 16384
net.netfilter.nf_conntrack_max = 1048576
kernel.pid_max = 4194304
net.core.default_qdisc = fq_codel
kernel.printk = 4 4 1 7
```

```
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
kernel.kptr_restrict = 1
kernel.sysrq = 176
vm.max_map_count = 1048576
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.all.rp_filter = 2
kernel.yama.ptrace_scope = 1
vm.mmap_min_addr = 65536
net.ipv6.conf.all.use_tempaddr = 0
net.ipv6.conf.default.use_tempaddr = 0
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
ubuntu@worker-01:~$
```

Install Kubernetes packages on Worker 1

```
ubuntu@worker-01:~$ sudo apt-get install -y apt-transport-https ca-certificates curl gpg
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.18.0-1ubuntu2.8).
curl set to manually installed.
gpg is already the newest version (2.4.8-4ubuntu3.1).
gpg set to manually installed.
Solving dependencies... Done
The following NEW packages will be installed:
  apt-transport-https
The following packages will be upgraded:
  apt ca-certificates libapt-pkg7.0
3 upgraded, 1 newly installed, 0 to remove and 53 not upgraded.
Need to get 2823 kB of archives.
After this operation, 42.0 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 libapt-pkg7.0 amd64 3.2.1ubuntu0.1 [1145 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 apt amd64 3.2.1ubuntu0.1 [1513 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 ca-certificates all 20260816ubuntu0.26.04.2 [138 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 apt-transport-https all 3.2.1ubuntu0.1 [26.0 kB]
Fetched 2823 kB in 0s (64.7 MB/s)
Preconfiguring packages ...
(Reading database ... 85427 files and directories currently installed.)
Preparing to unpack .../libapt-pkg7.0_3.2.1ubuntu0.1_amd64v3.deb ...
Unpacking libapt-pkg7.0:amd64 (3.2.1ubuntu0.1) over (3.2.0) ...
Setting up libapt-pkg7.0:amd64 (3.2.1ubuntu0.1) ...
(Reading database ... 85427 files and directories currently installed.)
Preparing to unpack .../apt_3.2.1ubuntu0.1_amd64v3.deb ...
Unpacking apt (3.2.1ubuntu0.1) over (3.2.0) ...
Setting up apt (3.2.1ubuntu0.1) ...
(Reading database ... 85427 files and directories currently installed.)
Preparing to unpack .../ca-certificates_20260816ubuntu0.26.04.2_all.deb ...
Unpacking ca-certificates (20260816ubuntu0.26.04.2) over (20260601~26.04.1) ...
Selecting previously unselected package apt-transport-https.
Preparing to unpack .../apt-transport-https_3.2.1ubuntu0.1_all.deb ...
Unpacking apt-transport-https (3.2.1ubuntu0.1) ...
Setting up apt-transport-https (3.2.1ubuntu0.1) ...
```

Add the Kubernetes repository key:

```
ubuntu@master:~$ kubectl get nodes
NAME        STATUS   ROLES           AGE   VERSION
master      Ready    control-plane   19m   v1.34.13
worker-01   Ready    <none>          23s   v1.34.13
ubuntu@master:~$
```

On worker-2

```
ubuntu@worker-02:~$ sudo apt install -y containerd
Installing:
  containerd

Installing dependencies:
  runc

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 56
  Download size: 37.9 MB
  Space needed: 142 MB / 17.2 GB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 runc amd64 1.4.0-0ubuntu1 [9829 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 containerd amd64 2.2.2-0ubuntu1.2 [28.1 MB]
Fetched 37.9 MB in 0s (91.1 MB/s)
Selecting previously unselected package runc.
(Reading database ... 85362 files and directories currently installed.)
Preparing to unpack .../runc_1.4.0-0ubuntu1_amd64v3.deb ...
Unpacking runc (1.4.0-0ubuntu1) ...
Selecting previously unselected package containerd.
Preparing to unpack .../containerd_2.2.2-0ubuntu1.2_amd64v3.deb ...
Unpacking containerd (2.2.2-0ubuntu1.2) ...
Setting up runc (1.4.0-0ubuntu1) ...
Setting up containerd (2.2.2-0ubuntu1.2) ...
```

Change systemcgroup to true

```
            CriuWorkPath = ''
            IoGid = 0
            IoUid = 0
            NoNewKeyring = false
            Root = ''
            ShimCgroup = ''
            SystemdCgroup = true

    [plugins.'io.containerd.cri.v1.runtime'.cni]
      bin_dir = ''
```

Disable swap

```
ubuntu@worker-02:~$ sudo swapoff -a
ubuntu@worker-02:~$
```

Configure kernel modules

```
kernel.apparmor_restrict_unprivileged_userns = 1
kernel.apparmor_restrict_unprivileged_unconfined = 1
kernel.core_pattern = core
net.ipv4.neigh.default.gc_thresh2 = 15360
net.ipv4.neigh.default.gc_thresh3 = 16384
net.netfilter.nf_conntrack_max = 1048576
kernel.pid_max = 4194304
net.core.default_qdisc = fq_codel
kernel.printk = 4 4 1 7
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
kernel.kptr_restrict = 1
kernel.sysrq = 176
vm.max_map_count = 1048576
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.all.rp_filter = 2
kernel.yama.ptrace_scope = 1
vm.mmap_min_addr = 65536
net.ipv6.conf.all.use_tempaddr = 0
net.ipv6.conf.default.use_tempaddr = 0
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
```

Install Kubernetes packages

Join Worker 2

On the master, generate a fresh join command:

```
ubuntu@worker-02:~$ sudo kubeadm join 10.30.1.170:6443 --token nghm7i.jwbqt2p9t5yn15kf --discovery-token-ca-cert-hash sha256:f236f4d904f83b6483aa03f2fcbd220f1bfc64c0988ad31
fe04b17f0116aafa1
[preflight] Running pre-flight checks
[preflight] Reading configuration from the "kubeadm-config" ConfigMap in namespace "kube-system"...
[preflight] Use 'kubeadm init phase upload-config kubeadm --config your-config-file' to re-upload it.
W1110 12:52:30.658092    7463 utils.go:69] The recommended value for "bindAddress" in "KubeProxyConfiguration" is: :::; the provided value is: 0.0.0.0
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/instance-config.yaml"
[patches] Applied patch of type "application/strategic-merge-patch+json" to target "kubeletconfiguration"
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[kubelet-start] Writing kubelet environment file with flags to file "/var/lib/kubelet/kubeadm-flags.env"
[kubelet-start] Starting the kubelet
[kubelet-check] Waiting for a healthy kubelet at http://127.0.0.1:10248/healthz. This can take up to 4m0s
[kubelet-check] The kubelet is healthy after 502.223057ms
[kubelet-start] Waiting for the kubelet to perform the TLS Bootstrap

This node has joined the cluster:
* Certificate signing request was sent to apiserver and a response was received.
* The Kubelet was informed of the new secure connection details.

Run 'kubectl get nodes' on the control-plane to see this node join the cluster.
```

run on master

```
ubuntu@master:~$ kubectl get nodes
NAME        STATUS   ROLES           AGE   VERSION
master      Ready    control-plane   29m   v1.34.13
worker-01   Ready    <none>          10m   v1.34.13
worker-02   Ready    <none>          39s   v1.34.13
ubuntu@master:~$
```

So Task 2 — Master + 2 Worker Nodes is complete.

Next task: Run an Nginx Pod

On the master, run:

```
ubuntu@master:~$ kubectl run nginx --image=nginx
pod/nginx created
ubuntu@master:~$ kubectl get pods -o wide
NAME    READY   STATUS    RESTARTS   AGE   IP              NODE        NOMINATED NODE   READINESS GATES
nginx   1/1     Running   0          22s   192.168.37.193   worker-02   <none>           <none>
ubuntu@master:~$ ^C
```